

International AS and A-level

Biology (9610)

BL02

Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL AS BIOLOGY (9610) BL02 - JANUARY 2025

The paper produced a wide range of marks with the highest mark being 68 (out of 75). The mean mark for the paper showed an increase of almost 4 marks from BL02 in the summer of 2024. Whilst there were many excellent answers across the paper, there was also a good number of responses that lacked clarity. Students should be reminded of the importance of reading the command words in questions carefully and following what is required.

Topics in which students demonstrated good knowledge included: the primary and secondary immune responses (**02.3**), knowledge of HIV and helper T-cells (**04.1** and **04.2**) and the enzymes involved in the digestion of proteins (**07.1**). Knowledge was less secure in the differences between cell division in eukaryotic and prokaryotic cells (**05.2**), and the differences between active and passive immunity (**02.1**). There were also misconceptions about the importance of the ventricles contracting from the base of the heart (**03.3**). Overall, the performance on mathematically based questions was mixed and generally not as strong as seen in previous series.

QUESTION 01

In question **01.1**, students were asked to give three practical precautions for setting up the potometer. The question was generally well answered, with around 70% of students achieving two or more marks. Around 35% of students achieved four or more marks. All the marking points (MP) were seen but the most common idea was to cut the plant shoot under water to prevent air entering the xylem. Some answers did not focus on the setting up part of the investigation, eg students must take repeat readings or just stated that a rubber bung must be used.

Just over 50% of students gained the mark in **01.2**. Where the mark was not given, it was typically for vague statements such as 'to add water back in the beaker'. Some students confused this investigation with a different practical, as the answers mentioned the use of the syringe to collect gases, or a volume of gas produced to be measured.

In question **01.3**, around 30% achieved all three marks. MP2 (describing the calculation for volume of water taken up) was the point that many students missed. Where this was the case, MP3 was still awarded for the idea of distance moved by the water level divided by time. Although **Figure 1** shows the water level in the capillary tube, many students wrote about movement of an air bubble. Whilst this was allowed, students should look at the labels in diagrams carefully and use the correct wording in their answers.

Question **01.4** was answered reasonably well, with just under 70% of students achieving one or more marks. Some students did not make a comparative statement which was required for this answer. Other answers included vague statements such as 'the plants are different species' or 'environmental conditions were different'.

QUESTION 02

Question **02.1** was answered reasonably well, with around 25% of students achieving one mark and around 45% gaining both marking points. Despite this, there were several misconceptions observed about active and passive immunity. In some cases, active immunity was described as the primary response, and passive as the secondary response. In other cases, active immunity was described as a response that requires energy and passive immunity requiring no energy. Several answers also showed some confusion between antigens and antibodies.

Just under 40% of students gained the mark in **02.2**. Where students didn't achieve the mark, it was usually for just giving units of volume. Arbitrary units and the number of antibodies per unit of volume were also not given credit.

Question **02.3** was answered well, with around 45% of students gaining full marks. There was evidence of good understanding in this topic area, with some detailed explanations for the differences in the primary and secondary responses. Some students made an error by stating that the secondary response takes more time. Others explained that the secondary response was greater because of antibodies still being present from the previous infection.

In question **02.4**, 50% of students correctly suggested surface proteins or antigens. Where students were not awarded the mark, it was typically for stating proteins unqualified.

Question **02.5** was not answered as strongly, with just under 40% of students getting the mark. Many referred to the mRNA being protected with no further qualification or just stating 'so that it can fuse with the membrane'. There were several references to preventing the virus from spreading in the body. It was evident that some students had not interpreted **Figure 3** correctly or had forgotten that the nanoparticles contained the viral mRNA, not the actual virus.

Just over 50% of answers got the mark in **02.6**, with B-cell, phagocyte, macrophage and antigen-presenting cell being common answers. The mark was not given for vague answers such as 'body cell' or 'white blood cell'.

QUESTION 03

Over 80% of students correctly named the right atrium in **03.1**. Incorrect answers included the right ventricle, left atrium or SAN.

Question **03.2** was not answered as well, with just under 40% of students gaining the mark. Many just stated 'to give a delay before the ventricles contract' or 'so the atria and ventricles don't contract at the same time', without any further details.

In a similar way, **03.3** was also not answered well, with around 25% gaining one mark, and just above 5% achieving full marks. Many answers just described the ventricles having a thicker muscle wall to withstand high pressures. Others described the ventricles needing to pump blood a longer distance thus needing more pressure, or that the thicker muscle at the base would give higher pressure. In some cases, students described blood flowing from the lower to upper chambers.

Just over 50% of students correctly calculated the heart rate in **03.4**. Where the mark was not given, it was usually for not reading from the graph in **Figure 5** accurately enough. In **03.5** students had to use the answer from the previous question to calculate the stroke volume. This question discriminated well, with around 55% achieving full marks and just over 20% getting one mark. Some students didn't convert dm^3 into cm^3 and others did the calculation the wrong way round.

Only 35% of students got full marks in **03.6** and just under 30% achieved one mark. Many students understood that point **X** would indicate the opening of the atrioventricular valve, but there was much confusion about the direction of blood flow. There was also some confusion between the semilunar and atrioventricular valves. Some students referred to pressure changes in the aorta, despite this not being shown in **Figure 5** or being required for the answer.

In question **03.7**, just over 20% achieved full marks and close to 40% gained one mark. Many just stated that the left ventricle pumps blood to the aorta which requires high pressure. A large number of answers referred to a thicker wall rather than thicker muscle layer. Some students just described that a thicker wall was needed to withstand the higher pressure in the ventricle.

QUESTION 04

Students showed good understanding of reverse transcriptase in question **04.1**, with 65% achieving full marks.

Question **04.2** was mostly answered well, with around 75% achieving at least one mark. Where marks were not awarded, it was usually for vague descriptions. For example, T-helper cells being important in coordinating the immune response, but with no further details.

In question **04.3**, students had to describe the trend in viral load between years 1 to 12. The question was answered well, with 65% gaining the mark. Despite this, many students did not read the question carefully or did not notice the time scale on the x-axis, and as a result, described the changes for the whole time period shown. Some described the trend for helper T-cell count instead of viral load. Others attempted to explain the trend in viral load, rather than focussing on the description of the trend.

Over 40% of students had the correct ideas for **04.4** and achieved full marks. Other answers gave either vague statements or measures not related to how a patient would be feeling. Examples included the amount of exercise/heart rate linked to different fitness levels, or T-cell count linked to different factors.

In **04.5**, around 40% of students correctly calculated the percentage change in viral load. Just over 20% achieved one mark for either giving the correct answer not to the nearest whole number or an answer to the nearest whole number but forgetting to multiply by 100.

Just over 80% of students gained one or more marks in **04.6**. All of these students gave the correct idea for MP1 and stated the decrease in number of helper T-cells or very low number of helper T-cells. The idea for MP2 was often not clearly expressed or given in vague statements eg there is a weaker immune response or just that pathogens are not killed.

QUESTION 05

Question **05.1** was answered very well, with nearly 75% of students gaining the mark. A misconception seen in a number of answers was meiosis for prokaryotes and mitosis for eukaryotes. There were also some references to 'binary fussion' which were not accepted.

In contrast, **05.2** was not answered as well, with less than 15% of students gaining full marks and around 40% achieving just one mark. Where marks were awarded, it was usually for references to prokaryotic cells in replicating plasmids or not using spindle fibres. MP1 was rarely seen, or answers didn't refer to replication of circular DNA and linear DNA, respectively. A misconception seen in several answers was that eukaryotic cells need two parents, but prokaryotic cells only need one parent (confusing the question for one comparing sexual and asexual reproduction).

In **05.3**, over 75% of students correctly worked out that there would be 48 divisions in the 24-hour period. Just over 40% gave the correct answer for full marks. There were several different errors, but the most common was to see evidence of 2^{48} but then forgetting to multiply this by the starting population.

Over 80% of students correctly described the main trend in **05.4** that as the concentration of vancomycin increased, the number of living bacteria decreased. Just a small proportion of students went on to describe how the number of living bacteria starts to level out from $50 \mu\text{g cm}^{-3}$. Some students just simply repeated the data or attempted an explanation.

In **05.5**, just over 40% of students gave a suitable limitation of the method. Many stated that the results were estimated numbers of bacteria and not actual numbers. Equally, there were many references to the investigation not being repeated. Answers referring to the length of the investigation were not given the mark, as it would not be relevant in this situation.

There were many good answers in **05.6**, with over 60% achieving at least one mark. The ideas of horizontal gene transmission and conjugation were seen in many answers. Where marks were not awarded, it was for answers such as 'through mutation' or 'from a common ancestor that passed the gene on to both species'.

QUESTION 06

Question **06.1** was not answered strongly, with many students just repeating parts from the question without adding further detail to actually answer the question. Just under 80% achieved at least one mark, but less than 5% achieved full marks. In nearly all the answers that gained a mark, it was usually for references to uncontrolled cell division or for the tumour suppressor gene being inactivated so unable to control cell division. In some of the better answers, MP2 was seen, but MP1 was only awarded in a very small proportion of answers.

06.2 was answered very well, with 75% of students correctly referring to the S-phase or interphase. Incorrect answers commonly referred to different stages of mitosis.

Students found **06.3** challenging, with just under 20% achieving at least one mark. It appeared that although many students had the general idea, the answers were not clearly

expressed. MP2 needed to be clear that only the non-senescent cells would become radioactive.

Students also found **06.4** quite challenging. Although 75% of students achieved at least one mark, only around 2% achieved full marks. Many answers showed that students had misinterpreted the graph in **Figure 7**. The graph showed the changes in concentration of p21 over 72 hours, but many students didn't refer to this aspect. Only the stronger answers referred to the fact that some cells in group 1 continued dividing, that the investigation was only done on isolated human cancer cells, or that the effects were only observed for 72 hours.

QUESTION 07

07.1 was answered well and successfully discriminated between students, with students demonstrating good understanding of the topic. Just over 50% of answers gained at least 4 marks and 35% achieved full marks. Some answers gave excellent descriptions of named enzymes including pepsin action in the stomach, which was mentioned in many answers. Where answers were not strong, it was typical to see vague references to proteases without giving any further details of protein digestion. Few students referred to dipeptidases and it was very rare to see mention of dipeptidases as membrane bound enzymes. Some students wrote at length about how amino acids are absorbed, without realising this wasn't required for the answer. A misconception observed in some answers was stomach acid causing the breakdown/hydrolysis of proteins.

Although not as strong as the previous part, **07.2** was still generally answered well. 75% of students achieved at least two marks, with 15% of these achieving full marks. References to the cholera bacteria producing a toxin were seen in most answers. MP3 was only seen in the stronger answers and MP2 – the idea of the toxin binding to receptors on the cell membrane of cells in the small intestine – was very rare.